

Gayatri Batta

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EDUCATION

Stony Brook University

New York, USA

Master's of Science in Artificial Intelligence, GPA:3.63/4

Aug 2024-May 2026

Courses: Analysis of Algorithms, Network Security Engineering, Deep Learning, NLP, Theory of AI

Dr. B.R. Ambedkar National Institute of Technology

Punjab, India

Major: Bachelor of Technology, Industrial and Production Engineering, GPA:8.67/10

Aug 2019-May 2023

Minor: Bachelor of Technology, Computer Science and Engineering

Courses: Data Analytics, OOP, RDBMS, Data Structures and Algorithms, Operating System

EXPERIENCE

Stony Brook University

New York, USA

Graduate Research Assistant - ECE Department - Dr. Petar M. Djuric

Jun 2025-May 2026

- Applied Recurrent Memory **Online Sparse Variational Gaussian Processes (OHSVGP)** with **HiPPO-based memory representations** to model **long-term dependencies** in patient calcium index trajectories using features like historical calcium index, serum calcium, calcitriol dosage, and IV supplements.
- Built a **probabilistic time-series forecasting pipeline** in **Python (PyTorch)**, integrating kernel methods and regression techniques to enhance predictive accuracy and support **personalized treatment planning**
- Reduced prediction error by **20%** compared to **vanilla GP and LSTM models** through effective long-term modeling.

Graduate Research Assistant - Neurosurgery Department - Dr. Sima Mofakham

Nov 2024-Jan 2025

- Processed **high-dimensional** MetaGloves sensor data to extract **movement patterns** for motor function analysis.
- Built **time-series pipeline** with **feature extraction** techniques to enable activity recognition from sensor signals.
- Delivered rehabilitation insights through **predictive modeling** and built interactive **visualization dashboards** using **Tableau** for clinical decision support.

Trianz Holdings Private Limited

Bangalore, India

Cloud and Data Engineer

Aug 2023-Aug 2024

- Created **Ansible playbooks** with **version control** for **microservices** deployment, reducing **40%** deployment errors.
- Built **ARIMA forecasting models** to predict cloud workloads, reducing infrastructure costs by **22%**.
- Automated ETL pipelines using **Python, AWS Lambda, and S3**; exposed metrics via **Flask APIs** with **Grafana** dashboards for monitoring.

Software Development and Automation Intern

Jan 2023-Jun 2023

- Implemented **infrastructure-as-code** using **Terraform** and **Ansible** across **AWS, Azure, and GCP**.
- Built automation recommendation engine with **PostgreSQL** and **FastAPI**, achieving **sub-200ms** response times.
- Built **CI/CD pipelines** with **automated unit and integration testing**, accelerating releases by **25%**.

PROJECTS

CodeGenAI – Multi-Agent Code Generation & Evaluation System | AutoGen, Python Feb 2026

- Built an AutoGen-based multi-agent system for code generation, Quality Assurance, and execution to **address** the critical **reliability and optimization gaps** in standard LLM code generation.
- Added **automated evaluation and logging** to assess correctness, performance, and reliability of generated code.
- Benchmarked** the workflow against LeetCode Weekly Contest Problems to mitigate **data contamination**.

MedifyAI- Health Condition Prediction Chatbot | LLM, NLP, Fine-tuning

Dec 2025

- Built MedifyAI, an intelligent chatbot that predicts health conditions using **Mistral-7B-Instruct-v0.2**.
- Applied **LoRA fine-tuning** with **4-bit quantization** and the **Paged AdamW optimizer** for optimal GPU efficiency.
- Improved prediction accuracy through **Cross-Entropy loss** optimization and **prompt tuning** for symptom diagnosis.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C++, SQL, PostgreSQL, Bash, Linux

Full Stack Development: React, Node.js, Flask, FastAPI, REST APIs, HTML, CSS

ML/AI: PyTorch, TensorFlow, Scikit-learn, LLMs, Fine-tuning, LangChain, Vector Databases, RAG, MLflow, Weights & Biases, ONNX, TensorRT, Streamlit

Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Plotly, Power BI, Grafana, Spark, Airflow, Streamlit

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Ansible, Git, JIRA, Confluence

AI-Assisted Development Tools: Cursor, GitHub Copilot, Claude Code